Engineering Mathematics - I · Unit_01 · Unit 1

DRAFT — UE26MA141A · Unit_01 · Unit 1 · Continuity and differentiability of a function of two independent variable

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Continuity and differentiability of a function of two independent variable

1 Continuity and differentiability of a function of two independent variable

Study focus: Continuity and differentiability of a function of two independent variable. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000002 · EV0000003

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today we'll be discussing continuity, history of partial derivative, definition of partial derivative, higher order partial derivative, and its geometric interpretation.

Key details

- In the previous session,.
- We defined what is the continuity of a function of single variable.
- A similar thing can be done for a function of two independent variables.
- We'll solve some examples so that the definition will be a little more clear to you.
- A function is continuous at a point.
- So geometrically it means that there is no hole or break in the graph.
- Unlike see for a function of single variable, what we said is a function is said to be discontinuous if there is a gap.
- But for a function of bivariate, I need to be in three-dimensional plane.
- I have x, y, and z axis.
- My surface would look something like this.
- But there is a hole.
- I'll show you how it looks.
- I can't call that as a break.
- A break can be considered only for a curve.
- But you need to be careful about something.
- A function is not continuous doesn't imply that the limit does exist.

- is not a condition, right?
- So that means even if the limit exists, the function need not be continuous.
- I know things are not that clear.
- If you observe carefully, as I go closer to 0, see if I come closer to 0, there is a folding that is happening.
- So that means there is actually a gap that is there at 0, 0.
- I need to check if this function is continuous or the curve must be very smooth.
- That's what I mean by continuous, right?
- How do I check if the function is continuous?
- I need to check the limit as xy.
- x y approaching to zero of this function 3x square y divided by x square plus y square this must be equal to f of 0 comma 0.
- That's the condition.
- But now is it satisfied is the question.
- For this function, what is the value of f of 0,0?
- If I substitute, I will get 3 into 0 into 0 divided by 0 plus 0, which is obviously in the form of 0 by 0, which is an indeterminate form.
- And no matter whatever the value I get in the left hand side, I cannot compare this with an indeterminate form.
- So this function is not continuous, whatever the function I have here is not continuous because the function is not defined at the origin.
- Limit might exist.
- But no matter whatever the value of this limit is, I cannot equate this to an indeterminate value.
- So function is not continuous because the value of the function is not defined at the origin.
- Is it clear?
- It's the same thing that we did yesterday.

Example or application from the lecture

- So let's take an example.
- Before we attempt to find the left hand side, let's check the value of f of 00.
- So I hope you understood the first example.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000002 · EV0000003 · EV0000004 · +1 source units in pack.json

Core explanation · 05:00–09:59

Let's see, what is this example now?

Key details

- You look at the graph carefully.
- This is the same function we took yesterday to explain the limits in the previous session, right?
- Limit for this.

- As, the limit does not exist, right?
- of this function, which is 2xy divided by 3x square plus y square.
- And we calculated along different paths we took y equal to 0 and then we took y is equal to x.
- x equal to 0.
- You take any path, the value that we get is different.
- Also the function, what is the value of the function at 0 comma 0?
- It will be 0 by 0, which is again an indeterminate form.
- Look at the graph a little bit carefully.
- At zero, there is actually a crunch or whatever you want to call it.
- It's like a boundary and then it suddenly drops down at 0, 0, and then of course at every other point the function exists.
- And as usual, as I mentioned in the previous lecture, it is very easy for us to check if the function is not continuous.
- If you want to check if the function is continuous, the best way to do that is look at the graph.
- Next, I want to go a little bit or talk a little bit about the history of partial differentiation.
- Who invented all these things?
- At least if not interested.
- Sometimes you want to curse people, right?
- You want to say why did he invent?
- Why did Newton invent so that I have to learn?
- Why did Jacobi do this?
- Why did Leibniz do this?
- but at least we should be grateful to few people who made our life much easier by inventing something that is very, very useful.
- It's only that you're hearing or it's been named differently, right?
- I hope you remember product rule.
- And all of you were very comfortable with product rule because things were very simple.
- Say I have so I have say x q.
- The product rule says that you keep the first function as it is, and you differentiate the second.
- Plus, you keep the second function as it is, and then you differentiate the first.
- And all of you were very comfortable when they define product rule.
- But did you question yourself that when I'm differentiating this, why am I not doing anything to this function here?
- It is just k cos x.
- You will not worry about whatever that constant is.
- The next time what you did, so let me just erase it.
- So the next time what you did is so what I did derivative of x cube into sine x.
- So in the first case I took this as K.
- And I treated that as k.
- So for the next term, that is the second term, what you did?
- You took sine x to be some constant.

Example or application from the lecture

- I took an example of a bivariate function.
- What does this product rule say is suppose I want to differentiate function, which is actually product of two functions.
- Let's take this.

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EV0000006 · EV0000007 · EV0000008 · +1 source units in pack.json

Core explanation · 10:00–14:59

And then so you treated that as K1 and you differentiated XQ which is 3x square.

Key details

- Is it clear whatever I'm talking?
- So partial differentiation also means the same thing.
- It's not something new.
- D of uv is u d v plus v d u, right?
- So Leibniz was the first person to introduce partial differentiation.
- In fact he in his letter to Lobithal he used the symbol delta.
- for partial derivative with respect to x.
- But later he didn't continue his work on partial derivatives.
- Legendary what he did is he actually defined the modern partial derivative.
- The partial derivative of a function.
- Then the partial derivative of this function is denoted by double.
- The symbol is not delta, this is dou.
- That's a notation.
- You have z, which is a function of more than one variable, and I'm going to differentiate that with respect to x by treating all other variables that are there with me as constant.
- That's what I'm going to do.
- Is the definition clear?
- So the re the very new concept of partial differentiation or the symbol that you're using is because of Jacobi.
- Jacobi actually reintroduced that symbol of Do, and then even today we use it.
- But there are other notations also for this.
- We will see how it is defined and, what it is and all.
- But the concept of partial differentiation was introduced by Leibniz, and then again it's like differentiating like it's like ordinary derivative.
- But you have to treat all other variables as constant and differentiate only the variable with respect to which we are finding the derivative.
- A partial derivative of a function of several variables is a derivative with respect to one of those variables with the other's held constant.
- And of course there are different notations.
- This is the most easier one for problems we preferably use this.
- You can find Zx, you can find ZY, and also you can find Z T.

- I am differentiating z like you differentiate any other function with respect to x, but with the constraint that you have to treat y and t as constant.
- As the definition is similar to the definition of the total derivative or derivative of a single variable function, all the properties, rules, and all those things that you used for the derivative of a function of single variable holds good here also.
- like product rule, quotient rule, chain rule, derivative of an implicit function, composite functions.
- All those things are applicable to partial derivatives also.
- I am taking bivariate, maybe I can do another one which is a multivariate function.
- Now, as soon as I ask you to differentiate, the first question that comes to your mind is with respect to what?
- I need to specify that.

Example or application from the lecture

- Suppose I have a function z is equal to f of x y.
- So let me take a simple example so that things will be a little more clear to you.
- Let's take a function which looks like say z is equal to.

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EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

Core explanation · 15:00–19:59

don't see any other thing, no matter how many variables you had here.

Key details

- Or I can add an e to the power of W to this and an Ln of say some P.
- But when I ask you to find ZX,.
- All these things doesn't matter for me because all of them will become constant.
- And I just need to differentiate x cubed, which is 3x square, and you retain all these things as it is because they are constant.
- Let me choose another color so that you can visualize properly.
- What will be ZP now?
- What all will be constant?
- You don't need to worry about any of these functions.
- All these functions are constant.
- And you just need to differentiate Ln of P, which will be 1 by P and the remaining things will come as it is.
- The next thing is let me choose another color.
- The other terms will remain as it is.
- As I said, all the rules of differentiation hold good for partial derivative also.
- But I'll discuss it more when we do problems.
- Again, when I say all the rules of differentiation holds good for partial derivative, that we found the first derivative and then we extend it to find the second derivative, third derivative.

- You can continue that.
- But there is one thing that is different in partial derivative.
- And Zx might look like sine y into 3x square, something like that.
- And I'll call that as Z Y.
- This so what will what will I get?
- I have to treat 3x square as constant, but I can differentiate sin y which will be cos y 4.
- With respect to itself is higher order, that's what we saw now x, zx and z y.
- But I can differentiate zx with respect to y and then z y with respect to x.
- And they are called the mixed partial derivatives.
- What does this mean is first you go in the same order as we are prefixing.
- First you calculate the derivative of z with respect to x and then with respect to y.
- So if I'm doing z y x, then that means I will differentiate first with respect to y and then with respect to.
- And then another x.
- The first thing you need to do is find zx.
- Zxy you differentiate with respect to X again.
- And then you finally you differentiate with respect to X again.
- That is, if the order of differentiation is immaterial.

Example or application from the lecture

- Suppose I have say I have zx.
- I'll continue with the same example that I did in the previous page.
- So suppose let's take an example.

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EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

Core explanation · 20:00–24:59

I would like to leave a note for you here to prove this Clydoids theorem.

Key details

- And the subscript notation, the proof is very simple.
- You don't need to do too much.
- Use the definition of limit definition of derivative and you can show that those two limits are equal.
- so next class we will see what is the geometry of this partial derivative.
- What geometrically what does the first derivative represent, second derivative represent.
- For a function of single variable, you already know that, right?
- The first derivative represents velocity, second derivative represents acceleration.
- But now we are in a higher dimension and we want to check what it represents.
- So that we'll take up in the next session.

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EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Continuity and differentiability of a function of two independent variable transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Continuity_and_differentiability_of_a_function_of_two_independent_variable__543f4d65e9b6ed52.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Continuity and differentiability of a function of two independent variable transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Continuity_and_differentiability_of_a_function_of_two_independent_variable__543f4d65e9b6ed52.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Continuity and differentiability of a function of two independent variable transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Continuity_and_differentiability_of_a_function_of_two_independent_variable__543f4d65e9b6ed52.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9985; only the JSON segmentation supplies the displayed timestamps.
Continuity and differentiability of a function of two independent variable transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Continuity_and_differentiability_of_a_function_of_two_independent_variable__543f4d65e9b6ed52.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00002	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00003	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...
SRC00004	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00005	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...

C Revision and retrieval practice